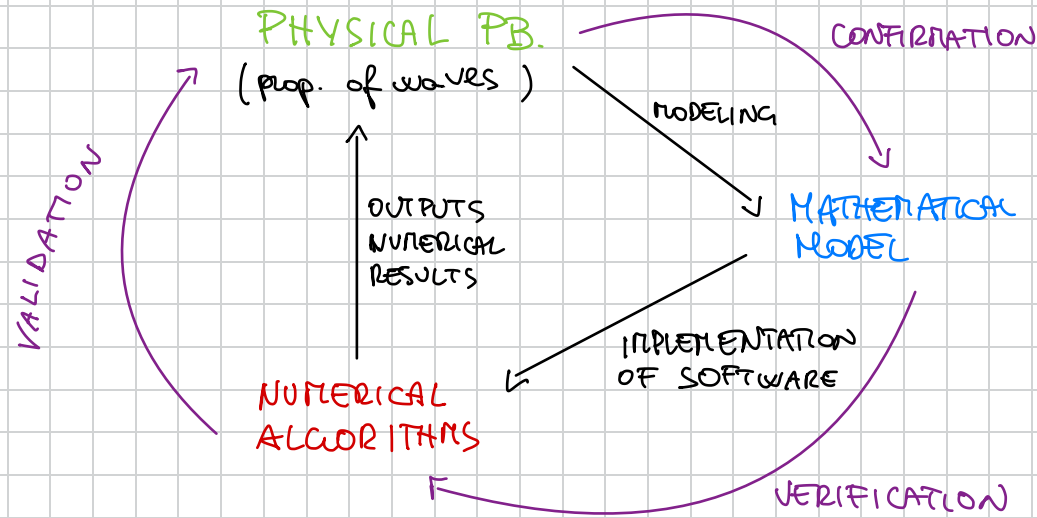
NUMERICAL MODELING & SIMULATION IN ACOUSTICS

Mathematematical modeling cycle



PB : WAVE PROPAGATION

NM : $F(\underline{x}, \underline{d}) = 0$

$\underline{x}$ solution (unknown)

$\underline{d}$ data

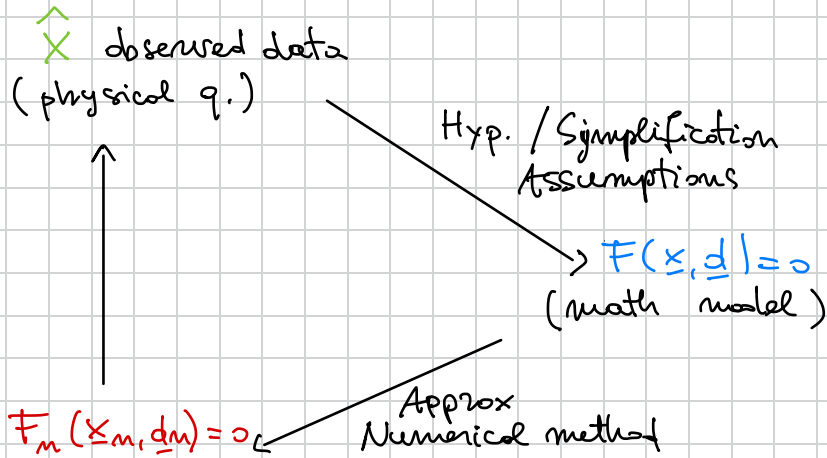
F equations

NA : $F_n(\underline{x}_n, \underline{d}_n) = 0$

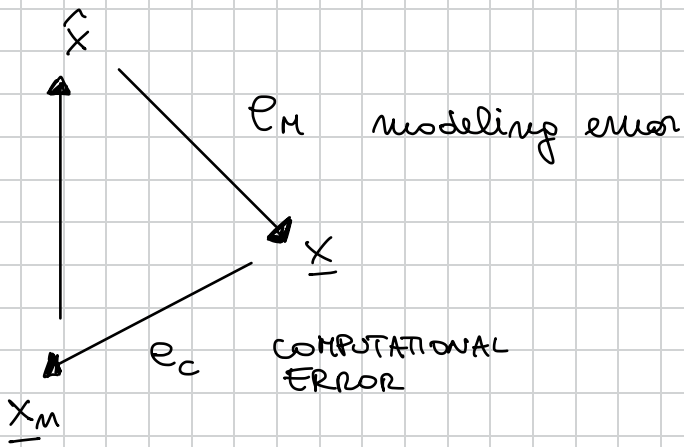
$\underline{x}_n$ approx solution

$\underline{d}_n$ approx. data

F_m approx eqs



$\hat{X}$, $\underline{x}$, X_m How are related to each other?



$e_C = e_D + e_A \longrightarrow$ ALGORITHMIC ERROR

$\hookrightarrow$ DISCRETIZATION ERROR

$$d(\underline{x}_m, \hat{x}) \leq d(\hat{x}, \underline{x}) + d(\underline{x}, \underline{x}_m)$$

d → DISTANCE

e → VALIDATION

e_M → CONFIRMATION

e_c → VERIFICATION

COMMON ASPECTS IN WAVE PROP

WAVE: transportation of energy through space and time

FINITE SPEED: that transport the energy

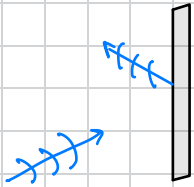
- **COMPRESSIONAL WAVES**: P-WAVES can travel through all media

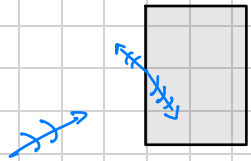
P WAVES → PARTICLE ↔

- **SHEAR WAVES**: S-WAVES

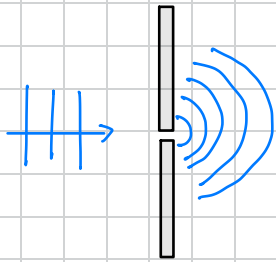
S WAVE → PARTICLE ↕

- **REFLECTION**
- **REFRACTION**

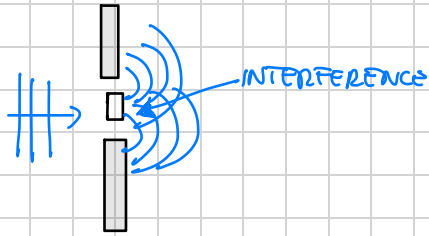




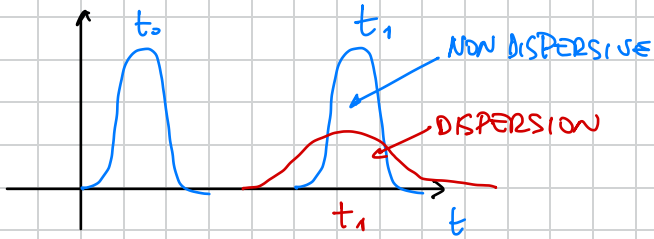
- DIFFRACTION



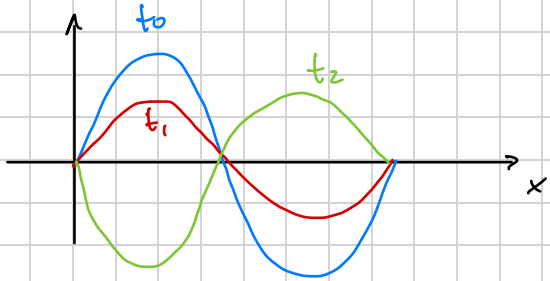
- INTERFERENCE



- DISPERSION



- STANDING WAVE

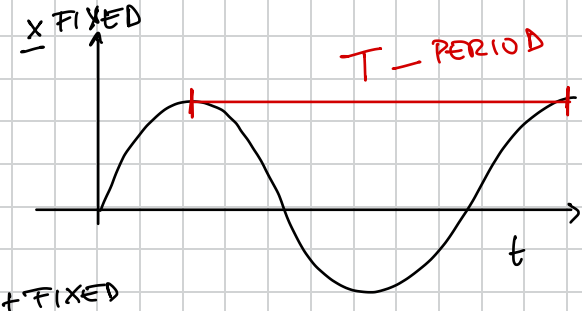


HARMONIC WAVES : $u(\underline{x},t) = A(\underline{x},t)\cos(\underline{k}\cdot\underline{x} - \omega t + \theta_0)$

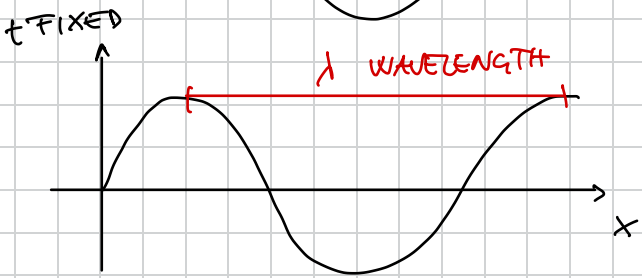
SHIFT IN PHASE

WAVE NUMBER

ANGULAR FREQ.



$$T = \frac{2\pi}{\omega} \qquad [s]$$



$$\lambda = \frac{2\pi}{k} \qquad [m]$$

FROM PHYSICS TO MATH : $(F(\underline{x}, \underline{d}) = 0\,?)$

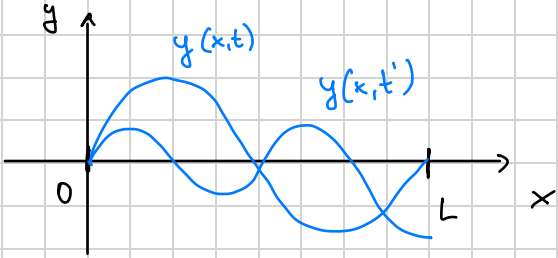
$F \to$ system of partial differential equation

$$F\left(\underline{d},\ u,\ \frac{\partial u}{\partial x},\ \frac{\partial u}{\partial y},\ \dots,\ \frac{\partial^2 u}{\partial y^2},\ \dots,\ \frac{\partial^2 u}{\partial t^2},\ \dots\right) = 0$$

WAVE EQ : $$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$$

(SIMPLEST MODEL FOR WAVE PROB.)

EXAMPLE : WAVE PROP. ON A STRING



L = length of the string

$y(x,t)$ SHAPE OF THE STRING FOR CERTAIN x AND CERTAIN TIME t

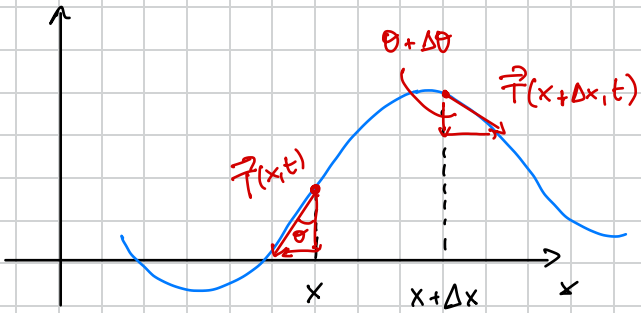
ASSUMPTION :

- SUPPOSE THE STRING TO HAVE CONSTANT MASS $\rho > 0$
- STRING CAN MOVE ONLY VERTICALLY
- TENSION $\vec{T}(x,t)$ ACTS ONLY TANGENTIALLY

EQUATIONS

- CONSERVATION LAWS
- CONSTITUTIVE EQs

NEWTON'S LAW $\vec{F} = m\vec{a}$. Translate into our settings.



$$\vec{F} = \vec{T}(x+\Delta x, t) - \vec{T}(x,t) \qquad T = |\vec{T}|$$

$$T_v = T \operatorname{sen} \theta$$

$$T \operatorname{sen}(\theta + \Delta\theta) - \underbrace{T \operatorname{sen}(\theta)}_{v(x,t)} = m\, a_v = \rho \Delta x\, a_v$$

$$v(x+\Delta x, t) - v(x,t) = \rho \Delta x\, a_v = \rho \Delta x \frac{\partial^2 y}{\partial t^2}(\bar{x}, t)$$

$$\bar{x} \in (x, x+\Delta x)$$

$$\lim_{\Delta x \to 0} \frac{v(x+\Delta x, t) - v(x,t)}{\Delta x} = \lim_{\Delta x \to 0} \rho \frac{\partial^2 y}{\partial t^2}(\bar{x}, t)$$

$$\boxed{\frac{\partial v(x,t)}{\partial x} = \rho \frac{\partial^2 y}{\partial t^2}(x,t)}$$

PHYS LAW

$$v(x,t) = T \operatorname{sen}(\theta)$$ VERTICAL TENSION

$$h(x,t) = T \cos(\theta)$$ HORIZ. TENSION

$$\frac{v(x,t)}{h(x,t)} = \operatorname{tg}(\theta) \Rightarrow v(x,t) = \operatorname{tg}(\theta)\, h(x,t)$$

$$\frac{\partial v}{\partial x} = \frac{\partial}{\partial x}(h \operatorname{tg}\theta) =$$

$$= \frac{\partial h}{\partial x} \operatorname{tg}\theta + h \frac{\partial}{\partial x}(\operatorname{tg}\theta)$$

$$\hookrightarrow \frac{\partial y}{\partial x}$$

$$= \cancel{\frac{\partial h}{\partial x}}^{\,0} \frac{\partial y}{\partial x} + h \frac{\partial^2 y}{\partial x^2}$$

$$\boxed{\frac{\partial v}{\partial x} = h(x,t) \frac{\partial^2 y}{\partial x^2}}$$

CONSTITUTIVE EQ !

$$\rho \frac{\partial^2 y}{\partial t^2} = h \frac{\partial^2 y}{\partial x^2}$$

$$\rho > 0 \implies \underbrace{\frac{\partial^2 y}{\partial t^2} = \overbrace{\frac{h}{\rho}}^{c^2} \frac{\partial^2 y}{\partial x^2}}_{\text{MATH. MODEL WAVE PROP. ON A STRING}}$$

$$c = \sqrt{\frac{h}{\rho}}$$

WAVE SPEED

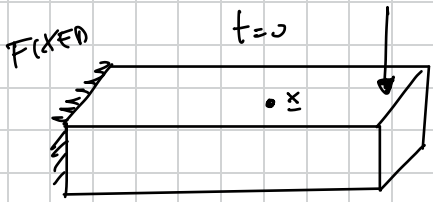
⊕ 2 INITIAL CONDITIONS : $y(x,0) = ?$

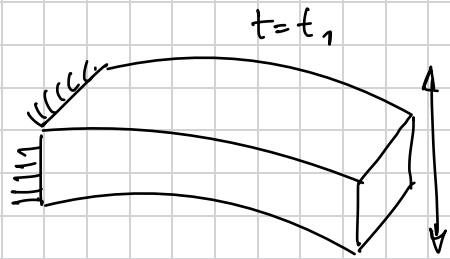
$$\frac{\partial y}{\partial t}(x,0) = ?$$

⊕ BOUNDARY CONDITIONS : $y(0,t) = \ldots$

$$y(L,t) = \ldots$$

EXAMPLE: WAVES IN SOLIDS





$$\underline{u}(\underline{x},t) = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}(\underline{x},t)$$

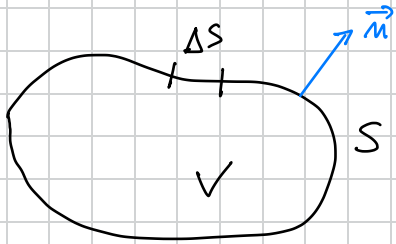
NEWTON'S LAW : $\vec{F} = m\vec{a}$

$$\vec{a}(\underline{x},t) = \frac{\partial^2 \vec{u}}{\partial t^2}(\underline{x},t)$$

$$\vec{F} = \vec{F}_{EXT} + \vec{F}_{INT}$$

$$\vec{F}_{EXT} = \vec{0}$$

$$\vec{F}_{INT} = ?$$



On the surface ΔS, σ (stress) exert a force in the normal direction called traction

$$\sigma \vec{n} \Delta S = \vec{T} \quad \text{(TRACTION)}$$

$$\underbrace{\int_S \sigma \vec{n} \Delta S}_{\text{TRACTION FOR THE WHOLE SURFACE}} = \int_V \nabla \cdot \sigma \, dV$$

DIVERGENCE THM

$$\vec{F}_{INT} = \nabla \cdot \sigma$$

$\nabla \cdot$ (divergence)

$$\Rightarrow \quad \boxed{\rho \frac{\partial^2 \vec{u}}{\partial t^2} = \nabla \cdot \sigma}$$

NEWTON'S LAW

(σ STRESS TENSOR)

$$\sigma = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{pmatrix} \quad (*)$$

CONSTITUVE EQ: HOOKE'S LAW (LINEAR ELASTIC ISOTROPIC SOLID)

$$\sigma = \lambda \, tr\left(\varepsilon(\vec{u})\right) + 2\mu \, \varepsilon(\vec{u})$$

1ST & 2ND LAMÉ PAR.

$\varepsilon(\vec{u})$ STRAIN TENSOR

$$\varepsilon(\vec{u}) = \frac{1}{2}\left(\nabla \vec{u} + \nabla \vec{u}^T\right)$$

SYM GRADIEN

tr = trace

$$tr(\sigma) = \sigma_{11} + \sigma_{22} + \sigma_{33}$$

$$\rho \frac{\partial^2 \vec{u}}{\partial t^2} = (\lambda + \mu) \nabla (\nabla \cdot \vec{u}) + \mu \Delta \vec{u}$$

CONST. EQ. + NEWTON

$$\cancel{\rho} \frac{\partial^2 \vec{u}}{\partial t^2} = \underbrace{\frac{(\lambda + 2\mu)}{\rho}}_{c_p^2} \nabla \cdot (\nabla \cdot \vec{u}) - \underbrace{\frac{\mu}{\rho}}_{c_s^2} \nabla \times \nabla \times \vec{u}$$

c_p^2 → COMPRESSIONAL (P) WAVE VELOCITY

c_s^2 → SHEAR (S) WAVE VELOCITY

EXAMPLE : WAVES IN FLUIDS (NAVIER-STOKES EQ.s)

$\vec{u}$ VELOCITY OF PARTICLES

p PRESSURE

MASS BALANCE $\quad \dfrac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{u}) = 0$

MOMENTUM BALANCE

$$\frac{\partial}{\partial t}(\rho \vec{u}) + \nabla \cdot (\rho \vec{u} \otimes \vec{u} - \sigma) = 0$$

$$(\vec{u} \otimes \vec{u})_{ij} = u_i u_j$$

$\sigma = -p I \quad$ PERFECT FLUID

$\Rightarrow$ INVISCID EULER EQs

FLUID UNDISTURBED $\quad \rho_0, \vec{u}_0, p_0 \quad$ &

SOUND WAVES ARE GENERATED BY LITTLE VARIATIONS AROUND THE EQUILIBRIUM.

$$\rho = \rho_0 + \rho' \quad , \quad \vec{u} = \vec{u}_0 + \vec{u}' \quad , \quad p = p_0 + p'$$

$p(\rho)$ TAYLOR EXP

$$p' = p - p_0 = \underbrace{\left(\frac{\partial p}{\partial \rho}\right)_s}_{c_0^2} (\rho - \rho_0) + \frac{1}{2}\left(\frac{\partial^2 p}{\partial \rho^2}\right)_s (\rho - \rho_0)^2$$

$$p - p_0 = c_0^2 (\rho - \rho_0)$$

MASS : $$\frac{\partial}{\partial t}\left(\cancel{\rho_0} + \rho'\right) + \nabla\cdot\left(\left(\cancel{\rho_0} + \rho'\right)\vec{u}'\right) = 0$$

NEGLECT 2ND ORDER TERM

MOMENTUM : $$\frac{\partial}{\partial t}\left(\left(\cancel{\rho_0} + \rho'\right)\vec{u}'\right) + \nabla\cdot\left(\cancel{\left(\rho_0 + \rho'\right)u'\otimes u'} + \left(\cancel{p_0} + p'\right)I\right) = 0$$

ρ_0, p_0 ARE CONSTANTS !

$$\frac{\partial}{\partial t}\left(\rho'\right) + \nabla\cdot\left(\vec{u}'\right) = 0$$

$$\frac{\partial}{\partial t}\left(\rho'\vec{u}'\right) + \nabla\cdot\left(p' I\right) = 0$$

$$\begin{cases} \dfrac{\partial \rho'}{\partial t} + \nabla\cdot u' = 0 & (1) \\ \dfrac{\partial}{\partial t}\left(\rho'\vec{u}'\right) + \nabla p' = 0 \end{cases}$$

$$\begin{cases} \dfrac{\partial}{\partial t}(1) \\ \nabla\cdot(2) \end{cases} \Rightarrow \frac{1}{c_0^2}\frac{\partial^2 p'}{\partial t^2} = \frac{\partial^2 p'}{\partial x^2}$$

$$\frac{\partial^2 \rho'}{\partial t^2} + \partial_t\left(\nabla\cdot u'\right) = 0$$

$$\nabla\cdot\underbrace{\left(\frac{\partial}{\partial t}\rho'\vec{u}' + \rho'\frac{\partial u}{\partial t}\right)} + \Delta p' = 0$$

TAYLOR EXP

$$\rho'\,\partial_t\left(\nabla\cdot\vec{u}\right) + \Delta p' = 0$$

EXAMPLE: ELECTROMAGNETIC WAVES

GAUSS LAW:

$$\begin{cases} \nabla \cdot \vec{E} = \dfrac{\rho_0}{\varepsilon_0} & (1) \\ \nabla \cdot \vec{B} = 0 & (2) \end{cases}$$

$\vec{E}$ electric field

ρ_0 CHARGE DENSITY

ε_0 PERMITTIVITY

$\vec{B}$ MAGNETIC FIELD

MAXWELL FARADAY LAW

$$\begin{cases} \nabla \times \vec{E} = -\dfrac{\partial \vec{B}}{\partial t} & (3) \\ \nabla \times \vec{B} = \mu_0 \left(\cancel{\vec{J}} + \varepsilon_0 \dfrac{\partial \vec{E}}{\partial t} \right) & (4) \end{cases}$$

μ_0 PERMEABILITY

J CURRENT FIELD

- $\vec{J} = \vec{0}$
- $\nabla \times (4) \Rightarrow \nabla \times \nabla \times \vec{B} = \mu_0 \varepsilon_0 \dfrac{\partial}{\partial t} \underbrace{(\nabla \times \vec{E})}_{(3)}$

$$\underbrace{\nabla \times \nabla \times \vec{B}}_{} = -\mu_0 \varepsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2}$$

$$\nabla \times \nabla \times \vec{B} = \nabla \big(\overbrace{\nabla \cdot \vec{B}}^{(2) \Rightarrow \nabla \cdot \vec{B} = 0} \big) - \nabla \cdot (\nabla \vec{B})$$

$$-\nabla \cdot (\nabla \vec{B}) = -\mu_0 \varepsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2}$$

$$\Delta \vec{B} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2}$$

WAVE EQ. FOR $\vec{B}$